DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO.4

Case study: Airline Management System

Mini Project: Centralized Club Management System

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Case Study: Airline Management System

Aim: Write code to implement Procedure and Function. for airline management system

Stored Procedure

A **Stored Procedure** is a precompiled set of SQL statements stored in the database that can be executed when needed.

- Can take input/output parameters
- Improves performance and reusability

Function

A **Function** is similar to a procedure but **returns a single value**.

- Must return a value
- Used inside SQL queries

```
mysql> USE airline_management;

Database changed

mysql> SHOW TABLES;

+-----------------------------+

| Tables_in_airline_management |

+-----------------------------+

| airline                     |

| airline_copy                |

| booking                     |

| flight                      |

| passenger                   |

+-----------------------------+

5 rows in set (0.04 sec)


mysql> select * from airline;

+------------+--------------+---------+--------------
+
```

```
| airline_id | airline_name | country |
founded_year |

+------------+--------------+---------+--------------
+

|          1 | Air India    | India   |        NULL |

|          2 | Indigo       | India   |        NULL |

|          3 | Spicejet     | India   |        NULL |

|          4 | Airindia     | India   |        NULL |

+------------+--------------+---------+--------------
+

4 rows in set (0.02 sec)


mysql> select * from booking;

+------------+-----------+--------------+-------------
--+---------+--------------+

| booking_id | flight_id | passenger_id |
booking_date | seat_no | travel_class |

+------------+-----------+--------------+-------------
--+---------+--------------+
```

```
|      1001 |      101 |        501 | 2024-02-01   | 12A     | Economy      |
|      1002 |      105 |        789 | 2023-09-08   | 14B     | Economy      |
|      1003 |      123 |        890 | 2025-07-07   | 13A     | Economy      |
+------------+-----------+--------------+-------------+---------+--------------+
3 rows in set (0.01 sec)

mysql> select * from flight;
+-----------+------------+---------+-------------+-------------+--------------+-------------+-----------+
| flight_id | airline_id | source  | destination | departure_time | arrival_time | flight_type | total_seats | status    |
+-----------+------------+---------+-------------+-------------+--------------+-------------+-----------+
|       101 |          1 | Mumbai  | Lucknow     | 10:00:00       | 12:00:00     | Domestic    |       180 | Scheduled |
|       102 |          2 | kolkata | nagpur      | 12:00:00       | 04:00:00     | Domestic    |       180 | Scheduled |
|       104 |          4 | mumbai  | chennai     | 06:00:00       | 09:00:00     | Domestic    |       180 | Scheduled |
+-----------+------------+---------+-------------+-------------+--------------+-------------+-----------+
3 rows in set (0.00 sec)


mysql> select * from passenger;
+--------------+----------------+--------+------+-----------+-------------+
| passenger_id | passenger_name | gender | age  | passport_no | nationality |
+--------------+----------------+--------+------+-----------+-------------+
|          501 | Rahul Mehta    | M      |   35 | P1234567    | Indian      |
|          678 | Rohit          | M      |   23 | P85666      | Indian      |
|          888 | Jiya           | F      |   46 | P45677      | Indian      |
+--------------+----------------+--------+------+-----------+-------------+
3 rows in set (0.00 sec)
```

Stored Procedures

1. Procedure to Get All Flights

```
DELIMITER //

CREATE PROCEDURE GetAllFlights()

BEGIN

   SELECT * FROM flight;

END //

DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetAllFlights()
    -> BEGIN
    ->     SELECT * FROM flight;
    -> END //
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetAllFlights();
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+------------+
| flight_id | airline_id | source  | destination | departure_time | arrival_time | flight_type | total_seats | status     |
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+------------+
|       101 |          1 | Mumbai  | Lucknow     | 10:00:00       | 12:00:00     | Domestic    |         180 | Scheduled  |
|       102 |          2 | kolkata | nagpur      | 12:00:00       | 04:00:00     | Domestic    |         180 | Scheduled  |
|       104 |          4 | mumbai  | chennai     | 06:00:00       | 09:00:00     | Domestic    |         180 | Scheduled  |
+-----------+------------+---------+-------------+----------------+--------------+-------------+-------------+------------+
3 rows in set (0.04 sec)

Query OK, 0 rows affected (0.09 sec)
```

2. Procedure to Get Passenger Details by ID

```
DELIMITER //

CREATE PROCEDURE GetPassengerById(IN pid INT)

BEGIN

   SELECT * FROM passenger

   WHERE passenger_id = pid;

END //

DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetPassengerById(IN pid INT)
    -> BEGIN
    ->     SELECT * FROM passenger
    ->     WHERE passenger_id = pid;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetPassengerById(501);
+--------------+----------------+--------+------+-------------+-------------+
| passenger_id | passenger_name | gender | age  | passport_no | nationality |
+--------------+----------------+--------+------+-------------+-------------+
|          501 | Rahul Mehta    | M      |   35 | P1234567    | Indian      |
+--------------+----------------+--------+------+-------------+-------------+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.04 sec)
```

3. Procedure to Count Bookings of a Passenger

```
DELIMITER //

CREATE PROCEDURE CountBookings(IN pid INT)

BEGIN

   SELECT COUNT(*) AS total_bookings

   FROM booking

   WHERE passenger_id = pid;

END //

DELIMITER ;
```

```
mysql> CALL CountBookings(501);
+----------------+
| total_bookings |
+----------------+
|              1 |
+----------------+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.03 sec)
```

4. Procedure to Insert Booking

```
DELIMITER //

CREATE PROCEDURE AddBooking(

   IN bid INT,
```

```
    IN fid INT,
    IN pid INT,
    IN bdate DATE,
    IN seat VARCHAR(10),
    IN class VARCHAR(20)
)
BEGIN
    INSERT INTO booking
    VALUES (bid, fid, pid, bdate, seat, class);
END //
DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE AddBooking(
    ->     IN bid INT,
    ->     IN fid INT,
    ->     IN pid INT,
    ->     IN bdate DATE,
    ->     IN seat VARCHAR(10),
    ->     IN class VARCHAR(20)
    -> )
    -> BEGIN
    ->     INSERT INTO booking
    ->     VALUES (bid, fid, pid, bdate, seat, class);
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL AddBooking(1004, 101, 678, '2025-01-01', '15A', 'Economy');
Query OK, 1 row affected (0.03 sec)
```

5. Procedure to Get Flights by Source City

```
DELIMITER //
CREATE PROCEDURE GetFlightsBySource(IN src VARCHAR(50))
BEGIN
    SELECT *
    FROM flight
    WHERE source = src;
```

END //

DELIMITER ;

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetFlightsBySource(IN src VARCHAR(50))
    -> BEGIN
    ->     SELECT *
    ->     FROM flight
    ->     WHERE source = src;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetFlightsBySource('Mumbai');
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
| flight_id | airline_id | source | destination | departure_time | arrival_time | flight_type | total_seats | status    |
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
|       101 |          1 | Mumbai | Lucknow     | 10:00:00       | 12:00:00     | Domestic    |         180 | Scheduled |
|       104 |          4 | mumbai | chennai     | 06:00:00       | 09:00:00     | Domestic    |         180 | Scheduled |
+-----------+------------+--------+-------------+----------------+--------------+-------------+-------------+-----------+
2 rows in set (0.02 sec)

Query OK, 0 rows affected (0.00 sec)
```

6. Procedure to Update Flight Status

DELIMITER //

CREATE PROCEDURE UpdateFlightStatus(

IN fid INT,

IN new_status VARCHAR(20)

)

BEGIN

UPDATE flight

SET status = new_status

WHERE flight_id = fid;

END //

DELIMITER ;

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE UpdateFlightStatus(
    ->     IN fid INT,
    ->     IN new_status VARCHAR(20)
    -> )
    -> BEGIN
    ->     UPDATE flight
    ->     SET status = new_status
    ->     WHERE flight_id = fid;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL UpdateFlightStatus(101, 'Delayed');
Query OK, 1 row affected (0.03 sec)
```

7. Procedure to Delete Booking

```
DELIMITER //

CREATE PROCEDURE DeleteBooking(IN bid INT)

BEGIN

   DELETE FROM booking

   WHERE booking_id = bid;

END //

DELIMITER ;
```

```
mysql> DELIMITER ;
mysql> CALL UpdateFlightStatus(101, 'Delayed');
Query OK, 1 row affected (0.03 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE DeleteBooking(IN bid INT)
    -> BEGIN
    ->     DELETE FROM booking
    ->     WHERE booking_id = bid;
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL DeleteBooking(1003);
Query OK, 1 row affected (0.01 sec)
```

8. Procedure with JOIN (Passenger + Flight Details)

```
DELIMITER //

CREATE PROCEDURE GetPassengerFlightDetails(IN pid INT)

BEGIN

   SELECT p.passenger_name, f.source, f.destination, b.seat_no

   FROM passenger p

   JOIN booking b ON p.passenger_id = b.passenger_id

   JOIN flight f ON b.flight_id = f.flight_id

   WHERE p.passenger_id = pid;

END //

DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetPassengerFlightDetails(IN pid INT)
    -> BEGIN
    ->     SELECT p.passenger_name, f.source, f.destination, b.seat_no
    ->     FROM passenger p
    ->     JOIN booking b ON p.passenger_id = b.passenger_id
    ->     JOIN flight f ON b.flight_id = f.flight_id
    ->     WHERE p.passenger_id = pid;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetPassengerFlightDetails(501);
+----------------+--------+-------------+---------+
| passenger_name | source | destination | seat_no |
+----------------+--------+-------------+---------+
| Rahul Mehta    | Mumbai | Lucknow     | 12A     |
+----------------+--------+-------------+---------+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.04 sec)
```

9. Procedure to Count Total Flights per Airline

```
DELIMITER //

CREATE PROCEDURE FlightsPerAirline()

BEGIN

   SELECT airline_id, COUNT(*) AS total_flights

   FROM flight

   GROUP BY airline_id;

END //

DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE FlightsPerAirline()
    -> BEGIN
    ->     SELECT airline_id, COUNT(*) AS total_flights
    ->     FROM flight
    ->     GROUP BY airline_id;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL FlightsPerAirline();
+------------+---------------+
| airline_id | total_flights |
+------------+---------------+
|          1 |             1 |
|          2 |             1 |
|          4 |             1 |
+------------+---------------+
3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

Functions

1. Function to Get Total Bookings of a Passenger

```
DELIMITER //

CREATE FUNCTION TotalBookings(pid INT)

RETURNS INT

DETERMINISTIC

BEGIN

   DECLARE total INT;


   SELECT COUNT(*) INTO total

   FROM booking

   WHERE passenger_id = pid;


   RETURN total;
```

END //

DELIMITER ;

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION TotalBookings(pid INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->
    ->     SELECT COUNT(*) INTO total
    ->     FROM booking
    ->     WHERE passenger_id = pid;
    ->
    ->     RETURN total;
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT TotalBookings(501);
+--------------------+
| TotalBookings(501) |
+--------------------+
|                  1 |
+--------------------+
1 row in set (0.01 sec)
```

2. Function to Get Passenger Age Category

DELIMITER //

CREATE FUNCTION AgeCategory(p_age INT)

RETURNS VARCHAR(20)

DETERMINISTIC

BEGIN

  DECLARE category VARCHAR(20);

  IF p_age < 18 THEN

    SET category = 'Minor';

  ELSEIF p_age BETWEEN 18 AND 60 THEN

```
        SET category = 'Adult';
    ELSE
        SET category = 'Senior';
    END IF;

    RETURN category;
END //
DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION AgeCategory(p_age INT)
    -> RETURNS VARCHAR(20)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE category VARCHAR(20);
    ->
    ->     IF p_age < 18 THEN
    ->         SET category = 'Minor';
    ->     ELSEIF p_age BETWEEN 18 AND 60 THEN
    ->         SET category = 'Adult';
    ->     ELSE
    ->         SET category = 'Senior';
    ->     END IF;
    ->
    ->     RETURN category;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT passenger_name, AgeCategory(age)
    -> FROM passenger;
+----------------+------------------+
| passenger_name | AgeCategory(age) |
+----------------+------------------+
| Rahul Mehta    | Adult            |
| Rohit          | Adult            |
| Jiya           | Adult            |
+----------------+------------------+
3 rows in set (0.02 sec)
```

3. Function to Get Flight Count of Airline

```
DELIMITER //

CREATE FUNCTION FlightCount(aid INT)

RETURNS INT

DETERMINISTIC

BEGIN

   DECLARE total INT;


   SELECT COUNT(*) INTO total

   FROM flight

   WHERE airline_id = aid;


   RETURN total;

END //

DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION FlightCount(aid INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->
    ->     SELECT COUNT(*) INTO total
    ->     FROM flight
    ->     WHERE airline_id = aid;
    ->
    ->     RETURN total;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT FlightCount(1);
+----------------+
| FlightCount(1) |
+----------------+
|              1 |
+----------------+
1 row in set (0.02 sec)
```

4. Function to Get Total Passengers in a Flight

```
DELIMITER //
CREATE FUNCTION TotalPassengers(fid INT)
RETURNS INT
DETERMINISTIC
BEGIN
    DECLARE total INT;

    SELECT COUNT(*) INTO total
    FROM booking
    WHERE flight_id = fid;

    RETURN total;
END //
DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION TotalPassengers(fid INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->
    ->     SELECT COUNT(*) INTO total
    ->     FROM booking
    ->     WHERE flight_id = fid;
    ->
    ->     RETURN total;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT TotalPassengers(101);
+----------------------+
| TotalPassengers(101) |
+----------------------+
|                    2 |
+----------------------+
1 row in set (0.00 sec)
```

5. Function to Get Airline Name from Flight ID

```
DELIMITER //

CREATE FUNCTION GetAirlineName(fid INT)

RETURNS VARCHAR(50)

DETERMINISTIC

BEGIN

   DECLARE a_name VARCHAR(50);


   SELECT a.airline_name INTO a_name

   FROM airline a

   JOIN flight f ON a.airline_id = f.airline_id

   WHERE f.flight_id = fid;
```

```
    RETURN a_name;

END //

DELIMITER ;
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetAirlineName(fid INT)
    -> RETURNS VARCHAR(50)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE a_name VARCHAR(50);
    ->
    ->     SELECT a.airline_name INTO a_name
    ->     FROM airline a
    ->     JOIN flight f ON a.airline_id = f.airline_id
    ->     WHERE f.flight_id = fid;
    ->
    ->     RETURN a_name;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT GetAirlineName(101);
+---------------------+
| GetAirlineName(101) |
+---------------------+
| Air India           |
+---------------------+
1 row in set (0.02 sec)
```

Mini Project: Centralized Club Management System

```
mysql> USE club_management;
Database changed
mysql> SHOW TABLES;
+--------------------------+
| Tables_in_club_management |
+--------------------------+
| attendance               |
| budget_request           |
| club                     |
| club_coordinator         |
| club_lead                |
| club_members             |
| event                    |
| reports                  |
| super_admin              |
+--------------------------+
9 rows in set (0.07 sec)

mysql> select * from attendance;
+---------------+-----------+----------+---------+
| attendance_id | member_id | event_id | status  |
+---------------+-----------+----------+---------+
|             2 |         2 |        1 | Absent  |
|             3 |         3 |        2 | Present |
|             4 |         4 |        3 | Present |
|             5 |         5 |        4 | Absent  |
+---------------+-----------+----------+---------+
4 rows in set (0.00 sec)

mysql> select * from budget_request;
+-----------+----------+----------------+---------+--
-------+------------+-----------------+
| budget_id | event_id | coordinator_id |
club_id | lead_id | submission |
approved_status |
+-----------+----------+----------------+---------+--
-------+------------+-----------------+
|         1 |        1 |            101 |       1 |     201
| 2026-04-01 | Pending        |
|         2 |        2 |            102 |       2 |     202
| 2026-04-05 | Approved       |
|         3 |        3 |            103 |       3 |     203
| 2026-04-07 | Pending        |
|         4 |        4 |            104 |       4 |     204
| 2026-04-10 | Approved       |
|         5 |        5 |            105 |       5 |     205
| 2026-04-12 | Pending        |
+-----------+----------+----------------+---------+--
-------+------------+-----------------+
5 rows in set (0.02 sec)

mysql> select * from club;
+---------+---------------+---------+----------------
+
```

```
| club_id | club_name     | lead_id |
coordinator_id |

+---------+---------------+---------+----------------
+

|      1 | Tech Club     |     201 |           101 |

|      2 | Cultural Club |     202 |           102
|

|      3 | Sports Club   |     203 |           103
|

|      4 | Music Club    |     204 |           104
|

|      5 | Art Club      |     205 |           105 |

+---------+---------------+---------+----------------
+

5 rows in set (0.01 sec)


mysql> selct * from club_coordinator;

ERROR 1064 (42000): You have an error in
your SQL syntax; check the manual that
corresponds to your MySQL server version
for the right syntax to use near 'selct *
from club_coordinator' at line 1

mysql> select * from club_coordinator;

+----------------+------------------+----------------
+----------+---------+

| coordinator_id | coordinator_name |
email          | password | club_id |

+----------------+------------------+----------------
+----------+---------+

|            101 | Anita            |
anita@gmail.com | pass     |       1 |

|            102 | Priya            |
priya@gmail.com | pass     |       2 |

|            103 | Rohit            |
rohit@gmail.com | pass     |       3 |

|            104 | Sneha            |
sneha@gmail.com | pass     |       4 |

|            105 | Vikas            |
vikas@gmail.com | pass     |       5 |

+----------------+------------------+----------------
+----------+---------+

5 rows in set (0.01 sec)


mysql> select * from club_lead;

+---------+-----------+------------------+----------
+---------+----------------+

| lead_id | lead_name | email            |
password | club_id | coordinator_id |

+---------+-----------+------------------+----------
+---------+----------------+

|     201 | Rahul     | rahul@gmail.com  |
pass     |       1 |           101 |

|     202 | Karan     | karan@gmail.com  |
pass     |       2 |           102 |

|     203 | Amit      | amit@gmail.com   |
pass     |       3 |           103 |

|     204 | Pooja     | pooja@gmail.com  |
pass     |       4 |           104 |

|     205 | Neeraj    | neeraj@gmail.com |
pass     |       5 |           105 |

+---------+-----------+------------------+----------
+---------+----------------+

5 rows in set (0.01 sec)


mysql> select * from club_members;
```

```
+------+--------+----------------+------------+-----
--------+----------+---------+

| m_id | m_name | m_email        |
m_password | event_roll  | domain   |
club_id |

+------+--------+----------------+------------+-----
--------+----------+---------+

|    1 | Aman   | aman@gmail.com  | 123
| Participant | AI       |       1 |

|    2 | Riya   | riya@gmail.com  | 123
| Volunteer   | Design   |       1 |

|    3 | Neha   | neha@gmail.com  | 123
| Participant | Dance    |       2 |

|    4 | Arjun  | arjun@gmail.com | 123
| Player      | Cricket  |       3 |

|    5 | Meera  | meera@gmail.com | 123
| Singer      | Vocal    |       4 |

|    6 | Kabir  | kabir@gmail.com | 123
| Artist      | Painting |       5 |

+------+--------+----------------+------------+-----
--------+----------+---------+

6 rows in set (0.00 sec)


mysql> select * from event;

+----------+----------------+------------+------------
+-----------+---------+---------+-----------+

| event_id | event_name     | start_date |
end_date   | report_id | lead_id | club_id
| budget_id |

+----------+----------------+------------+------------
+-----------+---------+---------+-----------+

|        1 | Hackathon      | 2026-04-01 |
2026-04-02 |      NULL |     201 |       1 |
NULL |

|        2 | Dance Fest     | 2026-04-05 |
2026-04-06 |      NULL |     202 |       2 |
NULL |

|        3 | Football Match | 2026-04-07 |
2026-04-08 |      NULL |     203 |       3 |
NULL |

|        4 | Music Night    | 2026-04-10 |
2026-04-11 |      NULL |     204 |       4 |
NULL |

|        5 | Art Exhibition | 2026-04-12 |
2026-04-13 |      NULL |     205 |       5 |
NULL |

+----------+----------------+------------+------------
+-----------+---------+---------+-----------+

5 rows in set (0.00 sec)


mysql> select * from reports;

+-----------+----------+---------+----------------+

| report_id | event_id | lead_id |
coordinator_id |

+-----------+----------+---------+----------------+

|         1 |        1 |     201 |            101 |

|         2 |        2 |     202 |            102 |

|         3 |        3 |     203 |            103 |

|         4 |        4 |     204 |            104 |

|         5 |        5 |     205 |            105 |

+-----------+----------+---------+----------------+

5 rows in set (0.00 sec)
```

```
mysql> select * from super_admin;

+----------+------------+-----------------+----------+----------------+

| admin_id | admin_name | email           | password | overall_budget |

+----------+------------+-----------------+----------+----------------+

|        1 | Admin1     | admin1@gmail.com | pass123  |      100000.00 |

|        2 | Admin2     | admin2@gmail.com | pass123  |      150000.00 |

|        3 | Admin3     | admin3@gmail.com | pass123  |      200000.00 |

|        4 | Admin4     | admin4@gmail.com | pass123  |      120000.00 |

|        5 | Admin5     | admin5@gmail.com | pass123  |      180000.00 |

+----------+------------+-----------------+----------+----------------+

5 rows in set (0.00 sec)
```

STORED PROCEDURE

1. Get all members of a specific club

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetMembersByClub(IN clubId INT)
    -> BEGIN
    ->     SELECT m.m_name, m.domain, c.club_name
    ->     FROM Club_members m
    ->     JOIN Club c ON m.club_id = c.club_id
    ->     WHERE m.club_id = clubId;
    -> END //
Query OK, 0 rows affected (0.11 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetMembersByClub(1);
+--------+--------+------------+
| m_name | domain | club_name |
+--------+--------+------------+
| Aman   | AI     | Tech Club |
| Riya   | Design | Tech Club |
+--------+--------+------------+
2 rows in set (0.03 sec)

Query OK, 0 rows affected (0.03 sec)
```

2. Insert new member

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE AddMember(
    ->     IN id INT,
    ->     IN name VARCHAR(50),
    ->     IN email VARCHAR(100),
    ->     IN pass VARCHAR(50),
    ->     IN role VARCHAR(50),
    ->     IN domain VARCHAR(50),
    ->     IN club INT
    -> )
    -> BEGIN
    ->     INSERT INTO Club_members
    ->     VALUES (id, name, email, pass, role, domain, club);
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL AddMember(7, 'Simran', 'simran@gmail.com', '123', 'Participant', 'AI', 1);
Query OK, 1 row affected (0.01 sec)
```

3. Get Events by Club

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetEventsByClub(IN cid INT)
    -> BEGIN
    ->     SELECT e.event_name, e.start_date, e.end_date
    ->     FROM Event e
    ->     WHERE e.club_id = cid;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetEventsByClub(1);
+------------+------------+------------+
| event_name | start_date | end_date   |
+------------+------------+------------+
| Hackathon  | 2026-04-01 | 2026-04-02 |
+------------+------------+------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

4. Update Member Domain

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE UpdateMemberDomain(
    ->     IN mid INT,
    ->     IN new_domain VARCHAR(50)
    -> )
    -> BEGIN
    ->     UPDATE Club_members
    ->     SET domain = new_domain
    ->     WHERE m_id = mid;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL UpdateMemberDomain(1, 'AI');
Query OK, 0 rows affected (0.01 sec)
```

5. Delete Member

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE DeleteMember(IN mid INT)
    -> BEGIN
    ->     DELETE FROM Club_members
    ->     WHERE m_id = mid;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL DeleteMember(6);
Query OK, 1 row affected (0.01 sec)
```

6. Get Attendance of Event

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetAttendanceByEvent(IN eid INT)
    -> BEGIN
    ->     SELECT m.m_name, a.status
    ->     FROM Attendance a
    ->     JOIN Club_members m ON a.member_id = m.m_id
    ->     WHERE a.event_id = eid;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetAttendanceByEvent(1);
+--------+--------+
| m_name | status |
+--------+--------+
| Riya   | Absent |
+--------+--------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

7. Get Budget Details by Club

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetBudgetByClub(IN cid INT)
    -> BEGIN
    ->     SELECT b.budget_id, b.approved_status, e.event_name
    ->     FROM Budget_Request b
    ->     JOIN Event e ON b.event_id = e.event_id
    ->     WHERE b.club_id = cid;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetBudgetByClub(1);
+-----------+-----------------+------------+
| budget_id | approved_status | event_name |
+-----------+-----------------+------------+
|         1 | Pending         | Hackathon  |
+-----------+-----------------+------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

FUNCTION

1. Count total members in a club

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION TotalMembers(clubId INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->
    ->     SELECT COUNT(*) INTO total
    ->     FROM Club_members
    ->     WHERE club_id = clubId;
    ->
    ->     RETURN total;
    -> END //
Query OK, 0 rows affected (0.11 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT TotalMembers(1) AS Total_Members;
+---------------+
| Total_Members |
+---------------+
|             3 |
+---------------+
1 row in set (0.07 sec)
```

2. Check if event is approved

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION CheckBudgetStatus(eid INT)
    -> RETURNS VARCHAR(20)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE status VARCHAR(20);
    ->
    ->     SELECT approved_status INTO status
    ->     FROM Budget_Request
    ->     WHERE event_id = eid;
    ->
    ->     RETURN status;
    -> END //
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT CheckBudgetStatus(1);
+----------------------+
| CheckBudgetStatus(1) |
+----------------------+
| Pending              |
+----------------------+
1 row in set (0.03 sec)
```

3. Count Events in a Club

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION TotalEvents(cid INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->
    ->     SELECT COUNT(*) INTO total
    ->     FROM Event
    ->     WHERE club_id = cid;
    ->
    ->     RETURN total;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT TotalEvents(1);
+----------------+
| TotalEvents(1) |
+----------------+
|              1 |
+----------------+
1 row in set (0.02 sec)
```

4. Get Club Name by Member ID

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetClubName(mid INT)
    -> RETURNS VARCHAR(100)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE cname VARCHAR(100);
    ->
    ->     SELECT c.club_name INTO cname
    ->     FROM Club_members m
    ->     JOIN Club c ON m.club_id = c.club_id
    ->     WHERE m.m_id = mid;
    ->
    ->     RETURN cname;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT GetClubName(1);
+----------------+
| GetClubName(1) |
+----------------+
| Tech Club      |
+----------------+
1 row in set (0.02 sec)
```

5. Count Present Members in Event

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION PresentCount(eid INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->
    ->     SELECT COUNT(*) INTO total
    ->     FROM Attendance
    ->     WHERE event_id = eid AND status = 'Present';
    ->
    ->     RETURN total;
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT PresentCount(1);
+----------------+
| PresentCount(1) |
+----------------+
|              0 |
+----------------+
1 row in set (0.02 sec)
```

6. Check Member Role

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetMemberRole(mid INT)
    -> RETURNS VARCHAR(50)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE role_val VARCHAR(50);
    ->
    ->     SELECT event_roll INTO role_val
    ->     FROM Club_members
    ->     WHERE m_id = mid;
    ->
    ->     RETURN role_val;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT GetMemberRole(1);
+------------------+
| GetMemberRole(1) |
+------------------+
| Participant      |
+------------------+
1 row in set (0.02 sec)
```

Conclusion

In this assignment, we successfully implemented advanced SQL concepts such as Stored Procedures and Functions using both the Airline Management System and the Centralized Club Management System. We created multiple stored procedures to perform operations like inserting, updating, deleting, and retrieving data efficiently. These procedures improved reusability and reduced redundancy in SQL queries.

We also implemented functions to perform calculations and return specific values, such as counting records, checking status, and retrieving specific information. Functions helped in simplifying queries and making them more modular and efficient.

Overall, this assignment enhanced our understanding of database programming, modular SQL design, and real-world database operations. It provided practical experience in improving performance, maintaining consistency, and organizing complex database logic effectively.